

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

The de Broglie wavelength of a particle is given by the equation :

$$\hat{\lambda} = h / p$$

where $\hat{\lambda}$ is the de Broglie wavelength , h is the Planck's constant (6.626×10^{-34} J s), and p is the momentum of the particle .

The kinetic energy (E) of the electron is given as 1 eV . We need to convert this energy into Joules :

$$E = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

We can use the equation :

$$E = (1/2)mv^2$$

to find the momentum (p) of the electron :

$$p = \sqrt{2mE}$$

where m is the mass of the electron (9.109×10^{-31} kg).

Now , we can calculate the de Broglie wavelength :

$$\hat{\lambda} = h / p$$

Substituting the values :

$$\hat{\lambda} = (6.626 \times 10^{-34} \text{ J s}) / (\sqrt{2 \times 9.109 \times 10^{-31} \text{ kg} \times 1.602 \times 10^{-19} \text{ J}})$$

$$\hat{\lambda} \approx 1.23 \times 10^{-7} \text{ m}$$

Converting the wavelength to centimeters :

$$\hat{\lambda} \approx 1.23 \times 10^{-7} \text{ cm}$$

Therefore , the de Broglie wavelength of the electron is 1.23×10^{-7} cm .

</think><answer> G . 1.23×10^{-7} cm </answer>